OG-PHL: PEP Simulation Results

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PEP Simulation Overview

- The Philippines Energy Plan (PEP) 2023 simulation models the economic and health impacts of energy sector transitions
- We compare a baseline scenario to the PEP 2023 scenario over a 50-year time horizon
- Key dimensions analyzed:
    - Energy sector parametrization (TFP, costs, investments)
    - Environmental outcomes (CO2e, PM2.5)
    - Health impacts (mortality rates, deaths averted)
    - Macroeconomic effects (GDP, capital, consumption, labor)
    - Fiscal implications (government spending, debt)
    - Distributional impacts (prices, income by ability group)

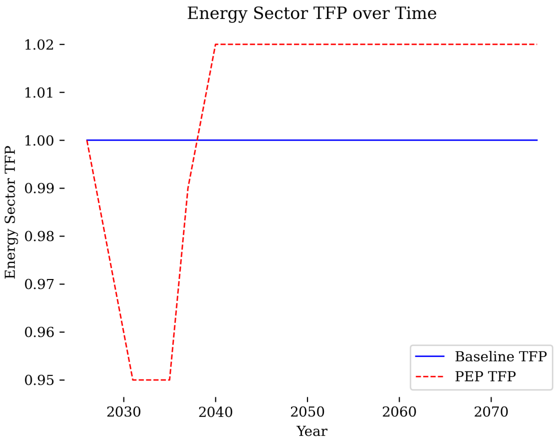
Input-Output Table for Multi-Industry Calibration

- Input-output table used for multi-industry calibration in PEP simulation

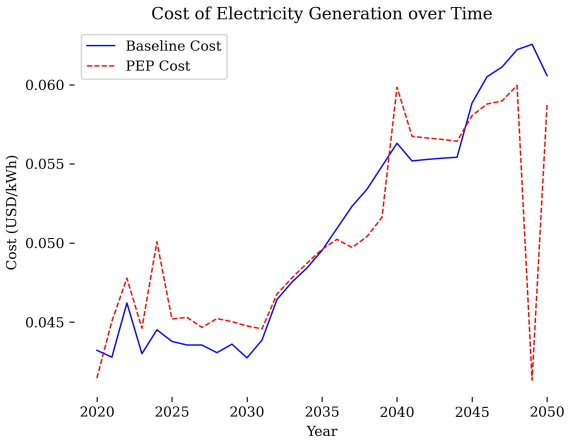
Input-Output Table used for Multi-Industry Calibration in PEP Simulation

	Natural Resources	Electricity	Construction, T
Food	0.120	0.000	
Energy and water	0.021	0.107	
Non-durables	0.043	0.006	
Durables	0.011	0.011	
Services	0.041	0.003	

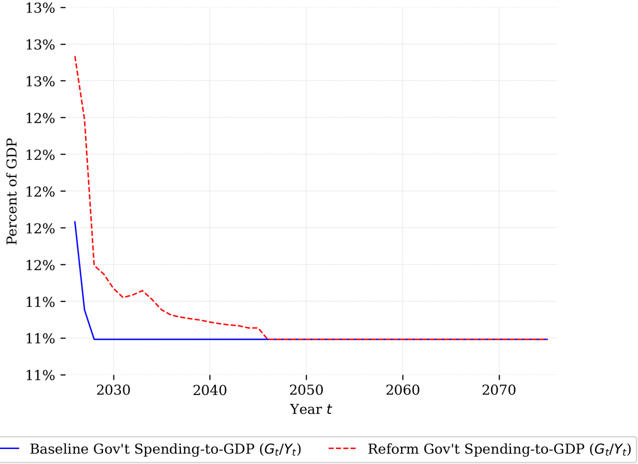
Energy Sector TFP over Time



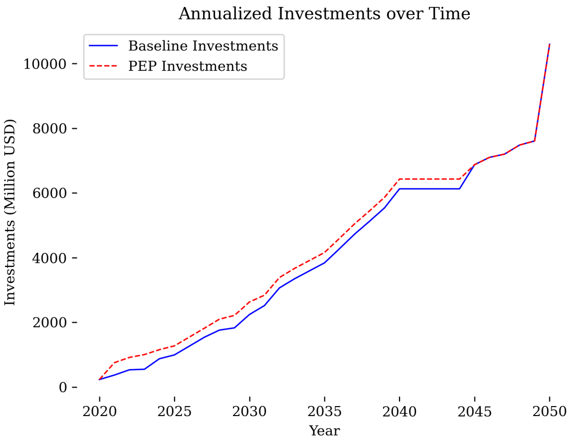
Cost of Electricity Generation over Time



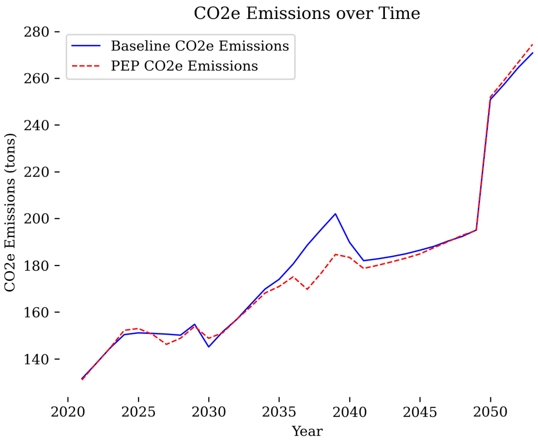
Government Spending over Time



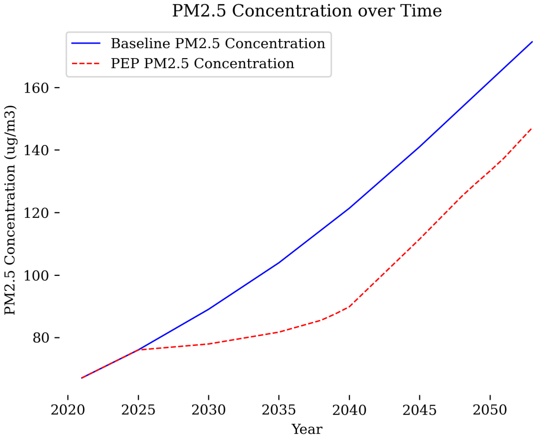
Annualized Investments over Time



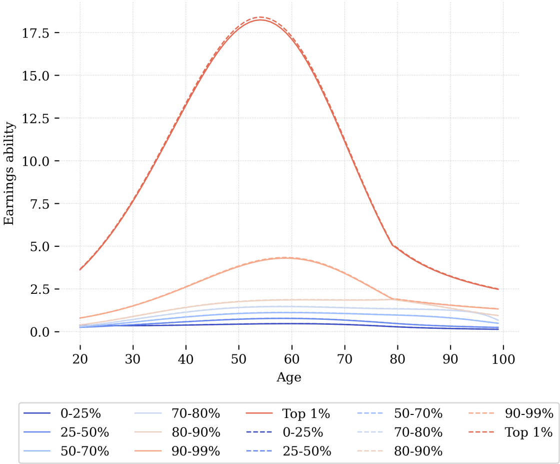
CO2e Emissions over Time



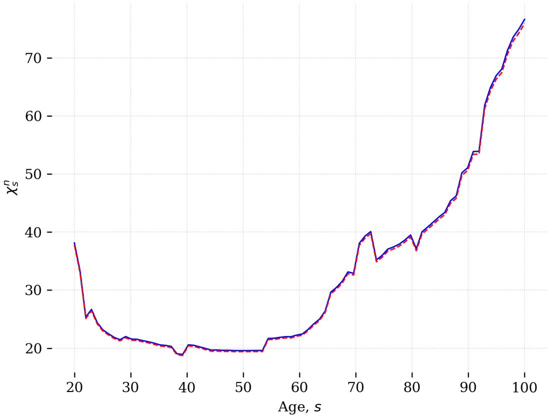
PM2.5 Concentration over Time



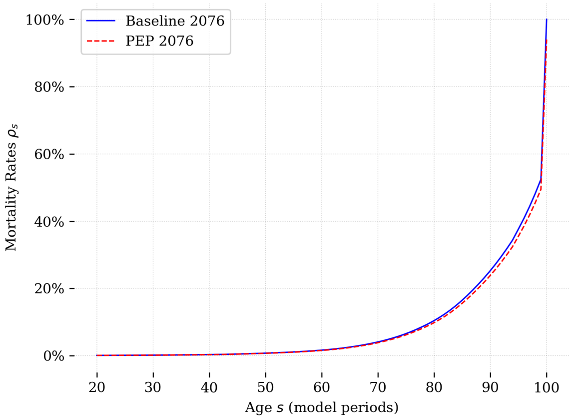
Ability Profiles in Baseline and PEP



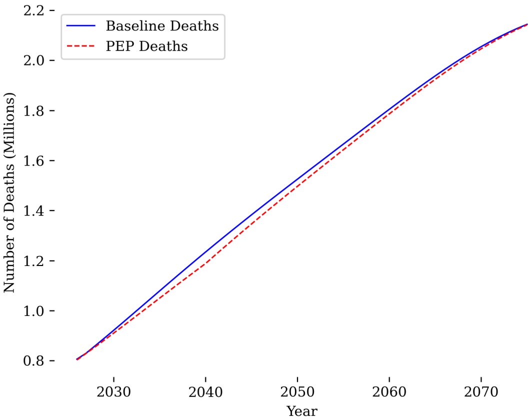
Chi_n Profiles in Baseline and PEP



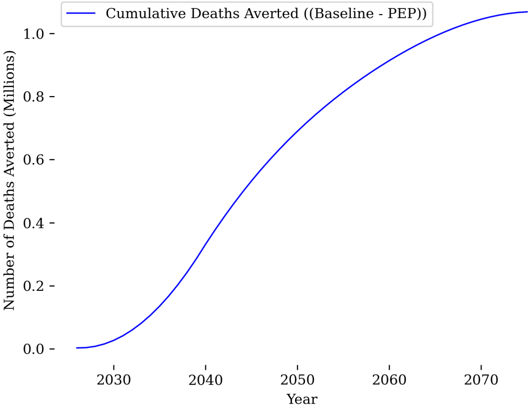
Mortality Rates in Baseline and PEP



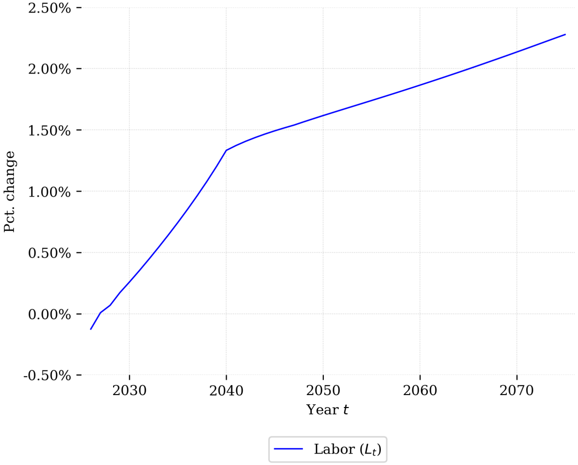
Annual Deaths over Time



Cumulative Deaths Averted over Time

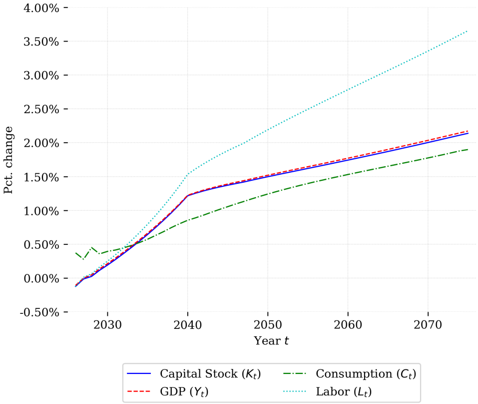


Labor Supply over Time

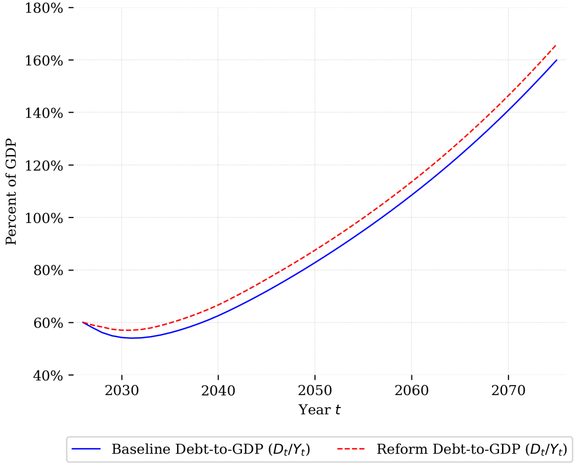


Aggregate Variables over Time

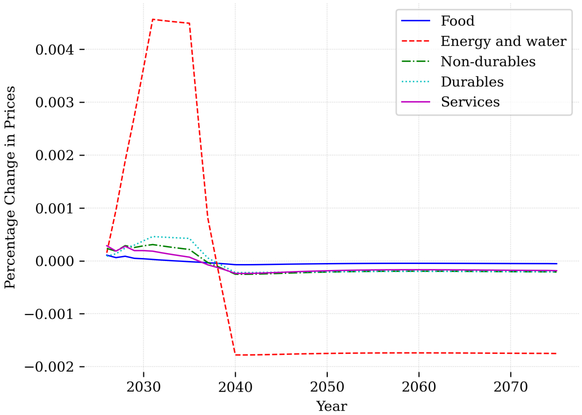
- Percentage changes in capital (K), output (Y), consumption (C), and labor (L)



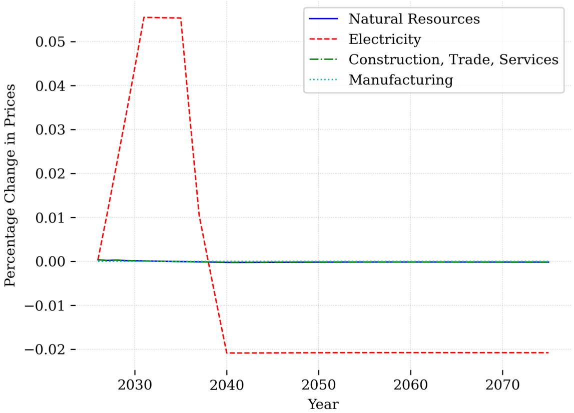
Debt-to-GDP Ratio over Time



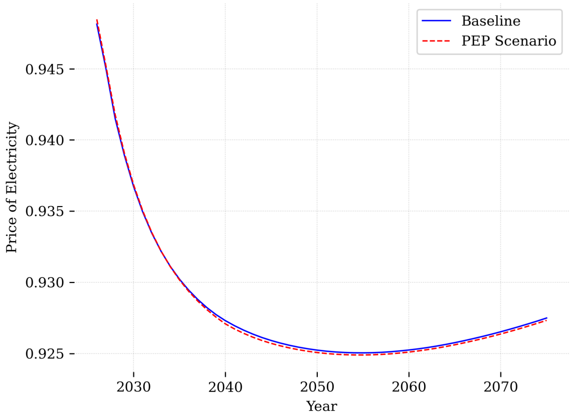
Percentage Change in Consumption Good Prices



Percentage Change in Production Good Prices

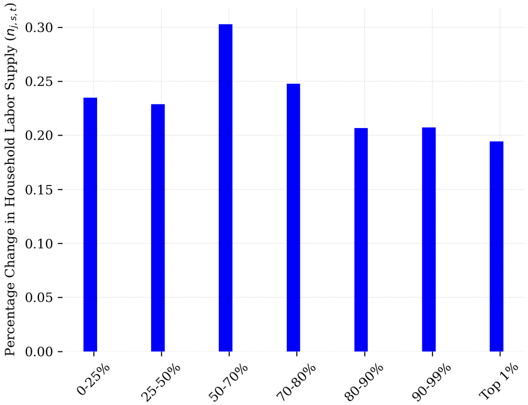


Price of Electricity



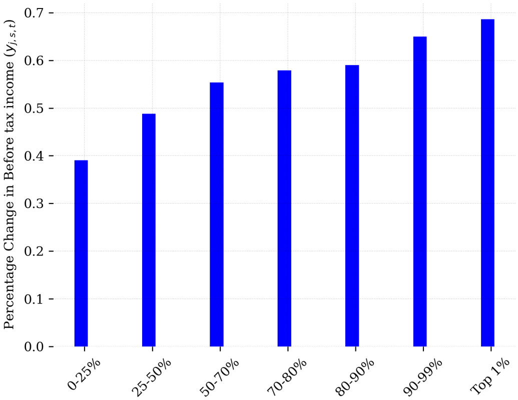
Percentage Change in Labor Supply by Ability Group

- Changes 10-20 years after policy implementation



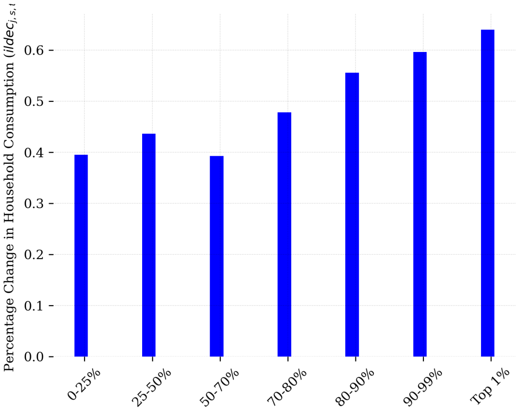
Percentage Change in Income by Ability Group

- Changes 10-20 years after policy implementation



Percentage Change in Consumption by Ability Group

- Changes 10-20 years after policy implementation



Key Findings

- Energy Transition:
  - PEP scenario shows increased energy sector TFP over time
  - Lower electricity generation costs in PEP scenario
  - Significant upfront investments required
- Environmental Benefits:
  - Reduced CO2e emissions
  - Lower PM2.5 concentrations
- Health Impacts:
  - Improved survival rates and lower mortality
  - Substantial cumulative deaths averted
  - Increased labor supply due to better health

Key Findings (continued)

- Macroeconomic Effects:
  - Positive impacts on capital, output, consumption, and labor
- Fiscal Implications:
  - Government spending increases for energy investments
  - Changes in debt-to-GDP ratio
- Distributional Effects:
  - Changes in relative prices across sectors
  - Electricity prices adjust over time
  - Differential impacts across ability groups